

The Solar System

Johanes Kepler¹

¹ Imperial Mathematician, Prague

1 Our Planetary Neighbourhood

1.1 The Sun and Its Planets

The Sun and its gravity-bounded bodies form the Solar System[1].¹ See NASA Solar System exploration. In increasing distance, the planets are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. The planets form two groups:

(I) Inner Planets

- † Mercury, Earth and Venus
- † they are comparatively
- † they are also called

(II) Outer Planets

- † Jupiter and Saturn are
- † Uranus and Neptune
- † All four outer

The inner planets are small, dense, and relatively close to the Sun, whereas the outer planets are much larger and travel along considerably wider orbits.

1.2 Portraits of the Solar System

Figure 1 shows the Sun and eight planets. 1 Compare Mars in Figure 1e with Jupiter in Figure 1f.

¹Sizes and distances are schematic

2 Kepler's Laws of Planetary Motion

Kepler's three laws describe planetary motion [3]. The first law describes the shape of planetary orbits, the second law describes the speed of planetary motion, and the third law relates the orbital period to the size of the orbit.

2.1 First Law: Elliptical Orbits

A planetary orbit is elliptical, with the Sun at one focus:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a \geq b > 0, \quad (1)$$

Here a and b are the semi-axes. For focal distance c ,

$$\begin{aligned} c^2 &= a^2 - b^2 \\ e &= \frac{c}{a} \end{aligned} \quad (2)$$

Here e is the eccentricity.

In polar form,

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}. \quad (3)$$

Equation (3) gives the orbital distance.

2.2 Second Law: Equal Areas in Equal Times

Equal times sweep equal areas:

$$\Delta t_1 = \Delta t_2 \implies \Delta A_1 = \Delta A_2 \quad (4)$$

Consequently, a planet moves

◇ faster when

◇ slower when

2.3 Third Law: Orbital Period and Distance

For planets orbiting one star,

$$T^2 \propto a^3 \quad (5)$$

Newtonian form:

$$T^2 = \frac{4\pi^2}{GM} a^3$$

Here M is the solar mass.

2.4 An Earth–Mars Comparison

Using Earth as reference, $a_E = 1AU$; $T_E = 1year$

3 The Outer Planets

The outer planets are Jupiter, Saturn, Uranus, and Neptune [2]. The first two are *gas giants*; the last two are *ice giants*.

Table 1 uses scientific and relativistic notation.²

Table 1: Orbital and Relativistic Parameters

Planets	Orbital Parameters		Relativistic Motion	
	a (AU)	e	$\epsilon = \frac{1}{ac^2}GM$	$\beta = \sqrt{\epsilon}$
Jupiter	5.204	0.0489	0.000	0.000
Saturn	9.582	0.0565	0.000	0.000
Uranus	19.201	0.0463	0.000	0.000
Neptune	30.047	0.0097	0.000	0.000

Table 1 compares the four giant planets.

References

- [1] Johannes Kepler. *Astronomia Nova*. Heidelberg, 1609.
- [2] National Aeronautics and Space Administration. Solar system temperatures, 2022.
- [3] National Aeronautics and Space Administration. Orbits and kepler’s laws, 2024.

²Values are rounded for typesetting practice.

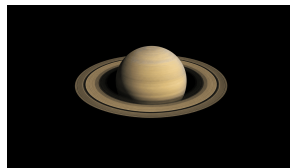
4 Conclusion



Figure 1: The Sun and its planets.



(a) Mars



(b) Saturn



(c) Jupiter

Figure 2: two panels

$$\mathcal{P} = \frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$